

Harvesting Insights: Unraveling Olive Dynamics and Climate Fluctuations through Regression and SHapley Additive explanations

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Abstract. Exploring the historical evolution of olive cultivation, this research investigates the complex relationship between climate variations and olive dynamics. This work presents a comprehensive analysis of olive influx prediction based on a meticulously compiled dataset derived from climate change studies in the Levant. The 269 data points in the dataset include a variety of variables, including age prior to present, precipitation levels, and monthly temperatures. Through strategic pre-processing and model evaluation, the proposed ensemble approach emerges as the best model for predicting olive influx, outperforming other regression methods. Using SHAP analysis, key climate factors such as temperature, precipitation, and tree age are identified, shedding light on their importance to olive dynamics. Furthermore, this study goes beyond model performance, delving into a historical dataset to investigate the complex relationship between climate fluctuations and olive production. This research not only advances knowledge of the dynamics of olives in Tyre, Lebanon, but also offers a useful framework for other areas facing climate-related challenges in olive cultivation. This work bridges the gap between agricultural sustainability and machine learning, providing crucial insights for generating resilience in olive-growing regions addressing climate-related issues

Keywords: Olive cultivation dynamics · Climate change impact · Ensemble regression model · SHAP analysis · Levant historical data

1 Introduction

Olives (*Olea europaea*) are silent participants in the constantly shifting tapestry of agriculture and climate, silent witnesses to the complex dance between the cycles of nature and human farming methods. The olive has a rich genetic diversity with over 1,200 cultivars, numerous wild trees, and many ancient cultivated

forms, making it an important agricultural species cultivated since ancient times across the old world [20][18][6]

Modern scientific technology is essential to olive cultivation in some Middle Eastern countries. In these areas, table olive production accounts for about 20% of the total olive acreage, with the remaining 80% being planted for olive oil production. Lebanon, situated in the Mediterranean region, boasts a rich heritage of olive cultivation, contributing to the diverse array of olive varieties and agricultural practices in the area[19]. Olives are an extremely significant historical and modern symbol of global ecological, cultural, and economic significance[5][9]. Across civilizations, they’ve been pivotal in culinary, economic, and environmental realms. Their cultivation is also intricately linked to the dynamic interplay of climatic fluctuations, which makes them an interesting research topic.[13]

The aim of this study is to delve into the dynamic relationship between climate fluctuations and olive production through the application of Regression and SHapley Additive exPlanations (SHAP) techniques. We aim to gain an understanding of the fundamental mechanisms governing olive dynamics and investigate the consequences for agricultural sustainability and climate resilience by dissecting the intricacies of this relationship.

Regression analysis is used in this study to investigate the long-term effects of climate variables on the dynamics of olive production. Additionally, SHAP is utilized to interpret and understand the predictive capabilities of the model, shedding light on the nuanced interactions between climate factors and olive influx.

It is critical for agricultural sustainability, climate change adaptation, and historical understanding to comprehend the complex relationship between climatic fluctuations and olive production[11][8]. Our research attempts to provide guidance for policy interventions, agricultural practices, and conservation efforts focused at protecting this priceless ecological and cultural resource by analyzing the fundamental drivers of olive dynamics.

The paper is organized as follows: Section 2 reviews related literature, Section 3 elucidates the research methodology, Section 4 presents experimental results including SHAP analysis, Section 5 provides a discussion, and Section 6 concludes the study while highlighting future directions.

2 Literature Review

Machine learning algorithms can help classify olive cultivars according to the chemical composition of their olive oil Skiada et al.[17]. Using chemometrics and multivariate analysis, several studies have successfully distinguished olive oil samples from various cultivars. Olive oil’s fatty acid makeup plays a major role in its classification, with certain fatty acids being recognized as important differentiators. It has also been demonstrated that other factors, including as acidity, chlorophyll, carotenoid, and sterol concentration, affect cultivar classification. Utilizing Deep learning methods such as convolutional neural networks further refine classification using data from ion mobility spectrometry and gas

chromatography, offering efficiency and objectivity surpassing traditional sensory analysis for olive oil classification

A few variables that affect the estimation of phenological stages in olive trees include the number of days and air temperature observed by N. Oses et al.[14]. The goal of the project is to use machine learning techniques to optimize phenological forecasts and find the base temperature parameter for heat unit accumulation in olive trees. Better model prediction was obtained by using a low base temperature, which is advantageous in the event of a rising environment. The study stressed how crucial it is to take into account satellite data and apply cross-validation procedures in order to guarantee that the final model can be applied to various years and geographical locations.

In Levant, olives are an important crop. Olive oil production faces a challenge from climate change. The work by Kaniewski, D et al.[7] compared geographic occurrences with data from contemporary climatology. This work used terrestrial sample 14 C dates for calibration. Based on increasing rate averages, the 2100 temperature anomaly was calculated. Restricted by the length of time records that are available for data analysis explained the anticipated impact of climate change on Lebanon's olive sector concluding that Levantine olive crops could decline due to climate change.

An early crop yield prediction machine learning model was created and embedded in a cloud-based web application for an olive grove in southern Spain by Cubillas, J.J. et al.[4]. The model made use of spatiotemporal training data, which included meteorological parameter data from a local weather station and yield data spanning eight years. The primary contribution of this study is the early yield forecast with absolute errors less than 20%, which is important for crop marketing and tillage investment decisions. With a web-based application, the farmer or farm manager may easily access and utilize the model.

An adaptable genetic approach has led to the development of the optimal deep learning model for identifying olive leaf diseases by Alshammari, Hamoud, et al.[1]. Utilizing parameterized transfer learning and intelligent data augmentation on an enhanced dataset, the model achieves an accuracy of approximately 96% for multiclass and 98% for binary classification, surpassing alternative methods. The MobileNet architecture, coupled with Rmsprop algorithms, emerges as the most effective CNN design for disease detection in olives. Additionally, a deep ensemble learning approach, combining vision transformer and convolutional neural network models, achieves high accuracy rates of around 97% for binary and 96% for multiclass classification.

Olive oil quality can be assessed by machine learning algorithms that are applied to spectrum data obtained from tiny spectrometers pointed out by Asharindavida F. et al.[2]. Miniaturized spectrometer experiments on olive oil quality assessment have produced encouraging findings, with classification rates as high as 98% and 96.73% attained. It has been discovered that applying pre-processing methods like wavelet analysis and normalization algorithms increases the classification models' effectiveness and accuracy. Furthermore, the examination of spectrum data obtained from small-scale spectrometers has shown promise in

distinguishing between 100% pure and contaminated olive oil. These results imply that, for the evaluation of olive oil quality, less expensive small spectrometers can yield results that are on par with more costly, non-portable spectrometers when paired with appropriate machine learning processes.

Assessing the efficacy of Explainable Artificial Intelligence (XAI) techniques such as SHAP, LIME, and CAM in offering transparent insights into AI models is imperative [12]. The significance of comparing evaluation metrics to identify the most trustworthy XAI techniques has been emphasised by recent studies. Metrics are important tools for evaluating the interpretability and dependability of these techniques, as well as highlighting their advantages and disadvantages. CAM (Class Activation Mapping) is another XAI technique that helps visualise the areas of an image that contribute to the model’s decision-making process. Examples of these techniques include SHAP and LIME, which are popular due to their effectiveness in explaining model predictions. Through the application of diverse metrics for assessing these techniques, researchers can improve the reliability and suitability of XAI for interpreting AI systems.

Enhancing the interpretability of AI models requires contrastive and counterfactual explanations [16]. A causality-based approach such as LEWIS introduces probabilistic contrastive counterfactuals to quantify direct and indirect influences on algorithmic decisions, whereas traditional methods concentrate on associational explanations. Post-hoc explanation techniques, such as SHAP, are widely used in the field of recommender systems to produce intelligible explanations while concealing underlying mechanisms. Strong axiomatic guarantees for producing contrastive and interpretable explanations can be obtained for complex inputs such as images by utilising the Shapley value framework and annotated interpretable features. These various strategies highlight how crucial it is to use SHAP to create thorough counterfactual and contrastive explanations in order to increase trust and transparency in AI decision-making processes.

3 Methodology

In this section, the road-map followed in the study is presented. The methodology is structured around pivotal stages: data collection, data pre-processing, model selection, and the application of an XAI framework to interpret the model predictions. The proposed methodology is summarized in Figure 1 which provides a simple and understandable overview of our strategy.

3.1 Data Collection and Pre-processing

The research utilizes a carefully compiled dataset sourced from the study “Climate change threatens olive-oil production in the Levant: past, present and future trends” [7]. This meticulously curated dataset includes 269 data points specifically designed to predict olive influx (grains cm⁻² yr⁻¹). The Lebanese olive dataset includes a wide range of features such as monthly temperatures, precipitation levels, age in years before present, percentiles, and more. Through

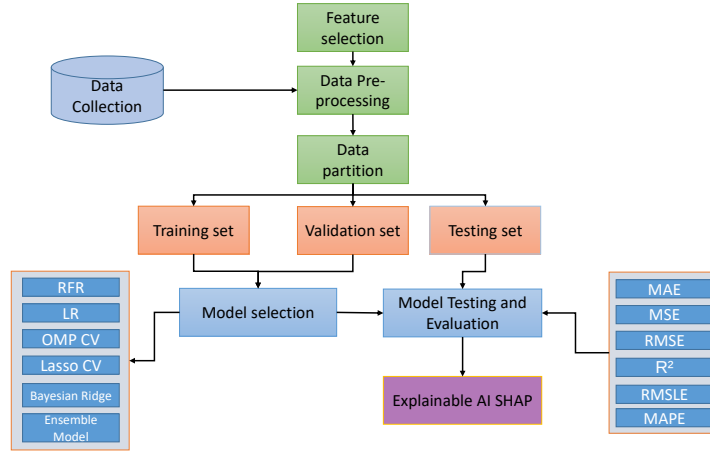


Fig. 1: Proposed methodology for olive influx based on climate factors.

strategic preprocessing, features have been reduced, condensing the dataset from 27 to 17 columns to enhance relevance and efficiency. Data preprocessing refers to the steps and techniques applied to raw data to transform it into a format that is more suitable for analysis or modeling[10][3]. To ensure robust model performance, we divided the dataset into training (80%) and test sets (20%), with an additional (20%) validation subset within the training data. This balanced distribution of 64:20:16 for training, testing, and validation ensures thorough preparation for analysis and predictive modeling. Though our dataset includes just 269 data points, each is rich in detail, spanning a historical range of 5,400 years. To maximize insights, we applied thorough preprocessing, reduced features strategically, and used a balanced train-test-validation split to ensure effective model training. Our chosen model handles small datasets well by reducing the impact of outliers and minimizing model variance. While we recognize the dataset's size as a limitation, its depth and historical scope allow us to confidently explore the long-term relationship between climate and olive dynamics. In Table 1, we present the data specifically partitioned for model training purposes.

Table 1: Train, Test and Validation split of the employed dataset.

Target Column	Number of Samples	Train	Test	Valid
Olive Influx (Grains $cm^{-2}yr^{-1}$)	269	172	54	43

3.2 Machine Learning Model Selection

Following data pre-processing, machine learning models were selected and evaluated for their predictive accuracy. Below are descriptions of all selected models.

Random Forest Regression (RFR): RFR, an ensemble learning method, constructs multiple decision trees during training and averages their predictions for regression tasks. The prediction of RFR for a given input vector can be expressed as:

$$\hat{Y} = \frac{1}{N} \sum_{i=1}^N f_i(x) \quad (1)$$

Where, N is the number of trees in the forest and $f_i(x)$ represents the prediction of the i -th decision tree.

Linear Regression (LR): LR aims to establish a linear relationship between a dependent variable (olive influx) and one or more independent variables.

$$\hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n \quad (2)$$

In Linear Regression, Y is the dependent variable (olive influx) $x_1, x_2, \dots, x_n$ are independent variable (environmental factor) and $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ are coefficients representing their effects

Orthogonal Matching Pursuit Cross-Validation (OMP-CV) Regression: OMP-CV is an iterative algorithm that selects a subset of relevant features while estimating coefficients simultaneously. The prediction $\hat{Y}$ of Orthogonal Matching pursuit cross-validation Regression for a given input vector x can be expressed as:

$$\hat{Y} = \sum_{i=1}^k \beta_i x_i \quad (3)$$

In OPM-CV Regression, β_i denotes the coefficient and x_i represents the i -th selected feature where k is the total number of selected features

Lasso Cross-Validation (LassoCV) Regression: LassoCV Regression combines Lasso regularization with cross-validation to select relevant features and estimate coefficients. The prediction $\hat{Y}$ of Lasso Cv Regression for a given input vector x can be expressed as:

$$\hat{Y} = x^T \beta \quad (4)$$

In Lasso CV, β represents the coefficient vector. While x^T denotes the transpose of the input vector x .

Bayesian Ridge Regression: Bayesian Ridge Regression provides a probabilistic framework for estimating coefficients and model uncertainty. The Bayesian Ridge Regression model can be represented as follows:

$$\begin{aligned}\hat{Y} &= X\beta + \epsilon \\ \beta &\sim \mathcal{N}(0, \lambda^{-1}\mathcal{I}) \\ P(\beta|D) &\propto P(D|\beta)P(\beta)\end{aligned}\tag{5}$$

Bayesian Ridge regression predicts $\hat{Y}$ using $X\beta + \epsilon$ incorporating regularization via γ and a prior distribution (β) . $P(\beta|D)$ captures data likelihood, ensuring robust predictions with uncertainty quantification

Ensemble Model (HuberRegressor + BayesianRidge): Ensemble modeling combines the predictive power of multiple models to improve overall performance. HuberRegressor offers robustness to outliers, while BayesianRidge provides uncertainty estimation, yielding a comprehensive approach. The ensemble model combines predictions from Huber regression $\hat{y}_{Huber}$ and Bayesian Ridge ($\hat{y}_{Bayesian}$) as follows:

$$\hat{y}_{ensemble} = \alpha \cdot \hat{y}_{Huber} + (1 + \alpha) \cdot \hat{y}_{Bayesian}\tag{6}$$

The ensemble prediction $\hat{y}_{ensemble}$ combines Huber Regression and Bayesian Ridge predictions using the blending parameter α , selected via cross validation.

3.3 Hyper-parameter Tuning

The best models were chosen once the pre-processing of the data was finished. Later, more hyper-parameters were adjusted to improve the performance of these chosen models. Table 2 presents the hyper-parameter configurations utilized in our models, aiming to achieve improved predictive outcomes.

In Table 2, Random Forest Regressor utilizes 100 estimators to build the model's more accurate performance. Linear Regression by default fits an intercept. Orthogonal Matching Pursuit CV and Lasso CV employ a cross-validation value of 5 to optimize their respective models. Bayesian Ridge utilizes a hyper-parameter `alpha_1` set to 1e-6 to control regularization. Within the Ensemble model, Huber Regressor's epsilon is set to 1.35, while Bayesian Ridge retains `alpha_1` of 1e-6. These hyper-parameters play critical roles in governing regularization strength, and model complexity, all of which influence the model's performance and generality.

3.4 Evaluation Matrices

After identifying the optimal models, their performance evaluation was conducted using six metrics: Mean Absolute Error (MAE), Mean Squared Error

Table 2: Hyper-parameter settings for the machine learning models utilized in this study.

Model	Hyperparameters
Random Forest Regresn	'n_estimators': 100
Linear Regression	'fit_intercept': True 'normalize': False 'copy_x': True 'n_jobs': None 'positive': False
Orthogonal Matching Pursuit CV	'CV': 5
Lasso CV	'CV': 5
Bayesian Ridge	'alpha_1': 1e-6
Ensemble (Huber+Bayesian)	'alpha_1'(Bayesian): 1e-6 'epsilon'(Huber): 1.35

(MSE), Root Mean Squared Error (RMSE), R-squared (R2), Root Mean Squared Logarithmic Error (RMSLE), and Mean Absolute Percentage Error (MAPE). The equations for these evaluation metrics are outlined below.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (7)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (8)$$

$$RMSE = \sqrt{MSE} \quad (9)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (10)$$

$$RMSLE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(1 + y_i) - \log(1 + \hat{y}_i))^2} \quad (11)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (12)$$

Where N is the number of samples, y_i is the actual values, $\hat{Y}_i$ is the predicted value and $\bar{Y}_i$ is the mean of the actual values.

3.5 Experimental Setup

3.6 Explainable Artificial Intelligence (XAI)

The goal of explainable artificial intelligence (XAI) is to provide users with a clear understanding of the factors influencing outcomes and the reasoning behind machine learning models' decisions. SHAP, a tool for explaining AI predictions by quantifying feature impact, was employed.

Shapley Additive exPlanations (SHAP): SHAP leverages game theory principles to offer comprehensive explanations for machine learning model predictions [15]. Mathematically, it can be expressed using the equation (eq 13).

$$\phi_i(x) = \sum_{S \subseteq \{1, \dots, P\} \setminus \{i\}} \frac{|S|! \cdot (P - |S| - 1)!}{P!} [f_i(S \cup \{i\}) - f_x(S)] \quad (13)$$

The SHAP value (ϕ) for a feature i in a specific prediction x is computed as the discrepancy between the model’s prediction when the feature i is included and the average prediction when i is excluded, aggregates across all feature combinations. This calculation elucidates the impact of individual features on model predictions, enhancing interpretability and facilitating insights into model behavior. The SHAP value (ϕ_i)(x) quantifies the impact of feature i in prediction x , considering all feature subsets except i . $f_x(S)$ represents the model’s prediction for feature set S among p total Features.

4 Experimental Results and Analysis

This section compares prediction results from different models using SHAP explanations. The ensemble model outperformed others, as detailed below.

4.1 Performance of Models

The final prediction results for all models are summarized in Table 3, derived from the testing data.

Table 3: Olive influx (grains cm-2 yr-1) prediction results from all employed models.

Model Name	MAE	MSE	RMSE	R^2	RMSLE	MAPE
Random Forest Regressor	10.64	206.67	14.38	0.94	0.02	7.59%
Linear Regression	6.57	104.88	10.24	0.97	0.00	4.47%
Orthogonal Matching Pursuit CV	5.96	93.06	9.65	0.97	0.00	3.85%
Lasso CV with Hyperparameters	8.98	177.36	13.32	0.95	0.01	6.09%
Bayesian Ridge Regressor with Hyperparameters	5.67	96.82	9.84	0.97	0.00	3.60%
Ensemble Model (HuberRegressor + BayesianRidge)	5.67	95.10	9.75	0.97	0.00	3.64%

In Table 3, when comparing olive influx prediction models using various metrics, the proposed ensemble model is consistently the best-performing model. It achieves a MAE of 5.67, matching the best result with Bayesian Ridge. The ensemble model shines in terms of MSE, boasting a score of 95.10, which surpasses all others. The Random Forest Regressor lags with the highest MSE of 206.67,

indicating its poor performance. While OMP CV slightly edges the ensemble in RMSE (9.65 vs. 9.75), both exhibit strong performance. All top models, including the ensemble, LR, OMP CV, and Bayesian Ridge, achieve a high R^2 of 0.97. The ensemble model stands out with a perfect RMSLE of 0.00, indicating precise error capture. Furthermore, with a MAPE of 3.64, the ensemble model surpasses OMP CV’s 3.85. Overall, across these metrics, the ensemble model consistently demonstrates superior performance, making it the optimal choice for Olive influx prediction.

Fig. 2: Model’s prediction output of the testing data

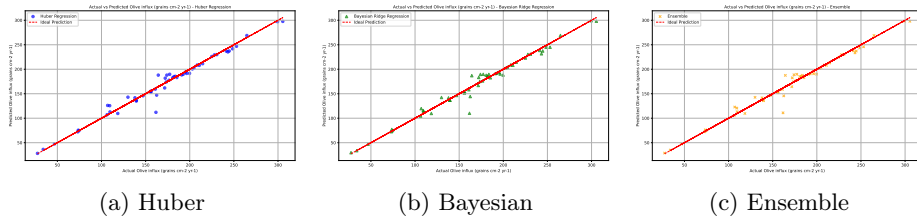


Figure 2 presents test results across three graphs: 2a for Huber regression, 2b for Bayesian ridge regression, and 2c showing the combined ensemble model, which outperforms both individual regressions.

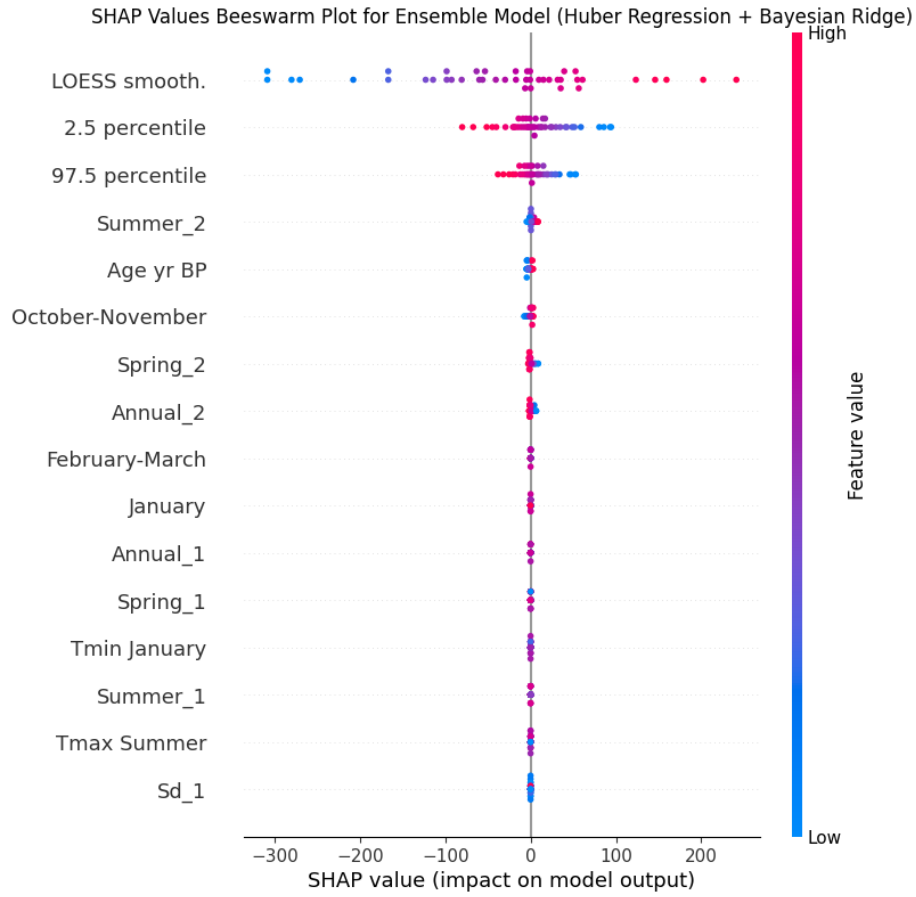
4.2 Using SHAP to Interpret Ensemble Model’s Predictions

The prediction results were analyzed using SHAP through beeswarm, waterfall, and force plots. Figure 3 presents a global explanation of the model through a beeswarm plot, illustrating the influence of each feature on predictions. According to SHAP analysis, ‘LOESS smooth.’, ‘2.5 percentile’, and ‘97.5 percentile’ emerge as the primary influencers. Higher values of these features positively impact predictions, while lower values exhibit a moderate negative influence. Notably, the target feature shows no distinct dependence on features other than the top influencers.

In Figure 4, waterfall plots for olive influx prediction illustrate the influence of features on explained instances. Three top influencers are identified. The baseline prediction $f(x)=37.836$ and the average prediction $E[f(x)]=180.535$ provide reference points for understanding feature contributions in the waterfall plot. In the first instance, ‘LOESS smooth.’ negatively impacts the prediction by -280.14, whereas ‘2.5 percentile’ and ‘97.5 percentile’ positively contribute to the prediction by 85.86 and 47.14, respectively. This demonstrates how different features have an impact on predictions.

In Figure 5, the SHAP force plot represents the model’s output for a specific instance as $f(x) = 31.66$. Plot range increases and decreases by 50 units, representing the range of predictions for various feature values. In color map’s

Fig. 3: SHAP beeswarm plot for the proposed ensemble model.



red zone, three features contribute positively to the prediction: 'summer_2' has the highest positive impact at 43.41, with its '97.5th percentile' at 52.61 and '2.5th percentile' at 22.22. In the blue zone, two features contribute negatively: 'LOESS smooth.' with a contribution of -37.87 and 'Age yr BP' with a contribution of -2369.0. This illustrates the importance of these features in influencing the model's prediction, with 'summer_2' having the most significant positive impact, while 'LOESS smooth.' and 'Age yr BP' exerting negative influences.

Fig. 4: SHAP waterfall plot for the proposed ensemble model.

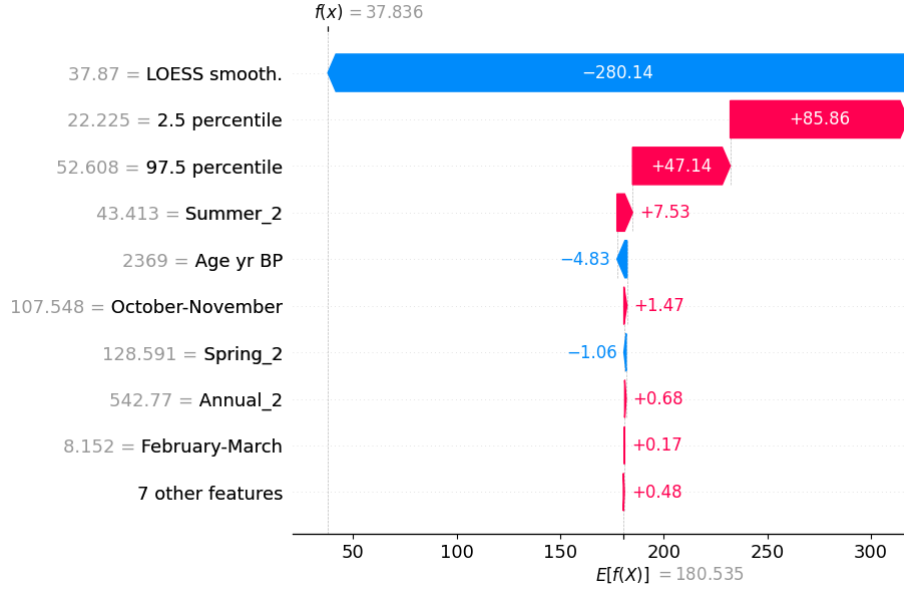
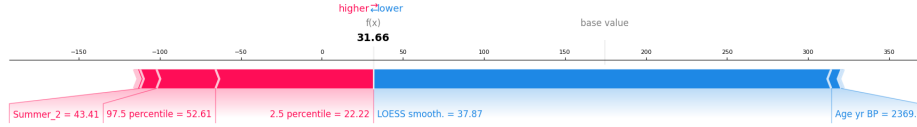


Fig. 5: SHAP force plot for the proposed ensemble model.



5 Discussion

The ensemble model demonstrates superior effectiveness in olive influx prediction, as evidenced by its performance surpassing that of other models, as depicted in Figure 2 and summarized in Table 3, which highlights its overall best performance among all models evaluated.

The SHAP analysis provides valuable insights into the factors influencing the model's predictions, consistently identifying certain features as top contributors such as temperature, precipitation, age year before present, and percentile, showcasing their strong correlation with the target variable, olive influx (grains cm-2 yr-1), and capturing temporal dependencies. Conversely, other features exhibit minimal influence on the predictions compared to these top contributors.

In the study investigating the relationship between climate and olive trees, a dataset spanning 5,400 years of olive tree dynamics in the ancient city of Tyre,

Lebanon, was analyzed. While our study focuses on Tyre, Lebanon, the findings are valuable for other Mediterranean and Levantine regions with similar climates and olive-growing practices. Tyre is representative of Mediterranean conditions, with seasonal temperature and rainfall patterns that are critical to olive cultivation. Although climate factors vary across regions, the climate-olive relationship identified here could apply to areas with similar environments. Expanding this research to other Mediterranean and semi-arid regions would allow for comparisons and help make our models more widely applicable. Significantly, before this investigation, there were no machine learning studies predicting olive influx in Tyre to the best of current knowledge, presenting an opportunity for the findings to benefit similar regions with comparable olive dynamics. A machine learning ensemble model employing regression techniques was developed to discern the climate factors most closely associated with olive influx (grains cm⁻² yr⁻¹) and to forecast their correlation and outcomes.

6 Conclusion and Future Work

The investigation into olive dynamics and climate fluctuations using regression analysis and SHapley Additive exPlanations (SHAP) has yielded significant insights. Through the investigation of the relationship between climate factors and olive influx, knowledge of Tyre, Lebanon's agricultural systems has been advanced. This information is essential for developing climate change adaptation plans and sustainable farming methods.

Future research endeavors could further delve into refining predictive models using advanced machine learning techniques, such as deep learning, to capture more complex relationships between climate variables and olive dynamics. Broadening the analysis to incorporate additional environmental factors like soil composition and water availability would enhance the depth of understanding in olive production dynamics. Furthermore, investigating the effects of other environmental factors on olive production and carrying out field research and experiments to validate model predictions would improve knowledge and help produce more precise predictions in comparable regions.

Data and Code Availability

The codes and data employed in this study can be found in the following GitHub repository. https://github.com/ShahriarAyon63/olive_influx_all

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